

RESEARCH ARTICLE

Cell culture process metabolomics together with multivariate data analysis tools opens new routes for bioprocess development and glycosylation prediction

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Abstract

Multivariate latent variable methods have become a popular and versatile toolset to analyze bioprocess data in industry and academia. This work spans such applications from the evaluation of the role of the standard process variables and metabolites to the metabolomics level, that is, to the extensive number metabolic compounds detectable in the extracellular and intracellular domains. Given the substantial effort currently required for the measurement of the latter groups, a tailored methodology is presented that is capable of providing valuable process insights as well as predicting the glycosylation profile based on only four experiments measured over 12 cell culture days. An important result of the work is the possibility to accurately predict many of the glycan variables based on the information of three experiments. An additional finding is that such predictive models can be generated from the more accessible process and extracellular information only, that is, without including the more experimentally cumbersome intracellular data. With regards to the incorporation of omics data in the standard process analytics framework in the future, this work provides a comprehensive data analysis pathway which can efficiently support numerous bioprocessing tasks.

KEYWORDS

CHO cell culture, glycosylation, metabolomics, multivariate analysis, partial least square regression

1 | INTRODUCTION

The biopharmaceutical industry has witnessed increasing diversification in the last decade,¹ with a tripled number of new biological licenses² and an estimated market volume above USD 390 billion by the year 2020.³ This results in an increasing pressure on the innovative path including drug discovery and process development with the

goal to offer increasingly case-specific and price-competitive therapeutics to patients. High-throughput (omics) technologies developed and optimized within the last decade have led to immense amounts of quantitative measurable metabolomics data and extensive knowledge of biological pathways⁴⁻⁹ paving the way for new process developments. Thereby, supportive quantitative tools for process monitoring, optimization, and control attain a central role.^{5,10} Given the complex and highly correlated nature of cell culture processes,¹¹⁻¹⁴ multivariate data-driven approaches are very effective to link product glycosylation information to process and metabolite information.^{12,15,16} Partial

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least square regression (PLSR) models demonstrate a good trade-off between prediction accuracy and interpretability.^{17,18} With regards to the high dimensionality and considerable noise level of both, process variables and critical quality attributes (CQAs), the combination of so-called PLS2 models with feature selection techniques proved to be advantageous.¹⁸ PLS2 models are capable of connecting the process to the product within a single model, while feature selection techniques^{18,19} enable to center variable importance interpretation on key features, such as operating parameters,²⁰ medium composition,²¹ and clone candidates.²¹ This procedure can then be used to sequentially define the design space of the key features fulfilling a certain product quality goal.²² However, such approaches have seldom been expanded to the extensive number of metabolic compounds present in extracellular and intracellular domains. The goal of this work is to apply such quantitative tools to process measurements while capturing the dynamics of several hundred extracellular and intracellular metabolites. Given the experimental effort required to generate metabolomics datasets, the major challenge of this work is the definition of a reproducible data analysis and interpretation framework based on a limited number of experiments. First, the elaborated prediction methodology for the glycan profile will be presented and then the obtained results will be discussed with regards to prediction accuracy and process understanding.

2 | MATERIALS AND METHODS

2.1 | Cell culture process and analytical procedure

In this work, a CHO-S derived clonal cell line expressing a human monoclonal IgG1 antibody was cultured in 3.5 L bioreactors according to a previously described fed-batch procedure.²³ Four differently designed parallel runs (referred to as Processes 1, 2, 3, and 4) with different media and main feed compositions were performed over 14 culture days as listed below in Table 1. Process performance (growth, viability, and product titer), extracellular glucose, lactate, and ammonium levels as well as the N-glycan profiles were analyzed without replicates according to previously reported methods.²³ The collected metabolomics samples from each bioreactor run were quenched in NaCl 0.9% (w/v) at 0.5°C. Supernatant and cell pellet samples were collected for exometabolome and intracellular metabolomics analysis, respectively, were flash-frozen in liquid nitrogen and stored at -80°C.

The subsequent sample preparation and the analysis of endo- and exometabolomes were performed as previously described.^{23,24}

2.2 | Description of fed-batch dataset

The analyzed three-dimensional dataset²⁵ comprises four cell culture runs (also referred to as processes) and four characteristic variable blocks which were evaluated at least at 12 time points throughout culture. The first block (IPC) contains six process variables, namely viable cell density (VCD), viability, the concentrations of glucose (Glc), lactate (Lac) and ammonium (NH₄), antibody titer as well as the variables representing the experimental design, are presented in Table 1. The second block represents the product quality attributes (CQA) containing 17 glycoforms. The third block contains the 117 extracellular metabolites and the last variable block consists of 168 different intracellular metabolites. The dataset was unfolded variable-wise,²⁶ resulting in 48 rows (4 runs times 12 time points), and as many columns as considered variables for the analysis. Some quality variables (CQAs) featured a considerable amount of missing data mostly due to quantities close to the limit of quantification (LOQ), so that one glycan (AO) with more than 22% of missing data was eliminated from the analysis, while for the other species the trimmed score regression imputation algorithm²⁷ and for the titer of the first days a logistic function-based approach was used.²⁸ In this work, three characteristic variable combinations will be considered for the description of the process: (a) *IPC* containing the six process variables and 4–6 designed conditions; (b) *IPC & Extra* additionally containing the extracellular variables (127 variables overall); and (c) *IPC & Intra* containing the intracellular measurements together with process variables (a total of 178 variables). The CQAs will be either described as 16 identified glycans or 5 characteristic groups, namely high mannose, afucosylated, fucosylated, galactosylated, and sialylated as previously described.²⁹

2.3 | Data analysis methodology

The statistical toolbox in Matlab (version R2017b, MathWorks, MA) was used for analysis. In a first step, univariate analysis of the time evolutions of the individual variables was performed. Additionally, principal component analysis (PCA)³⁰ was performed to gain information on the correlation structure of the CQAs as well as to determine

TABLE 1 Experimental design summary of the different medium and feed composition used in the four bioreactor runs regarding the asparagine (Asn) of the medium as well as the following feed composites: Asn, trace elements (TE), manganese (Mn), and galactose (Gal)

Process	Basal medium	Asn medium (–)	Asn feed (–)	TE feed (–)	Mn (μM)	Gal (mM)
1	Medium 1	1×	1×	1×	0	46
2	Medium 1	2.6×	2.7×	1.6×	5	0
3	Medium 2	1.8×	1×	2.9×	12	0
4	Medium 1	2.6×	2.7×	3.5×	5	0

Note: Feed supplements were added on days 3, 5, 7, and 10.

the multivariate batch evolution profiles. The data were auto-scaled prior to multivariate analysis by subtracting the mean of each column and dividing by the standard deviation.

For the generation of predictive models, the dataset was split into a calibration set consisting of the data of three runs, and a prediction set containing the information of the remaining run. This procedure was repeated four times so that each run could be predicted. Due to mathematical limitations of such rotation procedure, two almost unvaried variables (medium and Gn) were removed from the input variables. PLSR models were used to regress the CQAs (Y) based on the utilized process information (X). Model calibration was performed based on five-fold cross validation.³¹ Different PLSR models were compared, namely PLS1 models, where each CQA is regressed by a separate model, and PLS2 models where all CQAs are regressed simultaneously.^{15,17} Moreover, models created using all input variables were compared with models generated using reduced set of variables. Selection of the reduced variable sets was performed using the variable importance criterion (VIP) and the genetic algorithm method, initially developed by Leardi et al.³² and then adopted to PLS2 by Sokolov et al.¹⁸ The prediction quality was quantified for the calibration and the prediction either by the variance explained (Q^2) or the root mean square error (RMSE).³³

3 | RESULTS AND DISCUSSION

3.1 | Univariate analysis

Distinct performance, metabolite, and glycosylation responses were observed in the four 3.5 L bioreactor processes, which are displayed for several variables in Figure 1. Cell growth was comparable in all four processes (1–4 and) over the first 144 hr. Process 3 reached the peak viable cell density at 216 hr amounting to 24.4×10^6 viable cells/ml. Processes 2 and 4 exhibited slightly lower peak cell densities. Moreover, both were characterized by pronounced viable cell density reductions, as the production phase progressed. In contrast, for Process 1 reached peak cell density 2 days earlier and was lower (18.8×10^6 VC/ml). The recombinant protein titer of Process 1 was high despite the reduced cell growth. Its titer at harvest (2.7 g/L) was in the same range as Process 3 (2.8 g/L). The product titer yielded by Processes 2 and 4 was substantially lower, amounting to 1.9 and 2.3 g/L, respectively. All cultures showed the highest lactate level at 66 hr of about 2 g/L. A rapid metabolic shift to lactate consumption took place in Process 1, thus leading to a strong decline in extracellular lactate. Beyond 168 hr, lactate remained below the LOQ. Process 3 showed slower lactate consumption, reaching the LOQ at 232 hr. Processes 2 and 4, first followed the trend of Process 3, whereby the

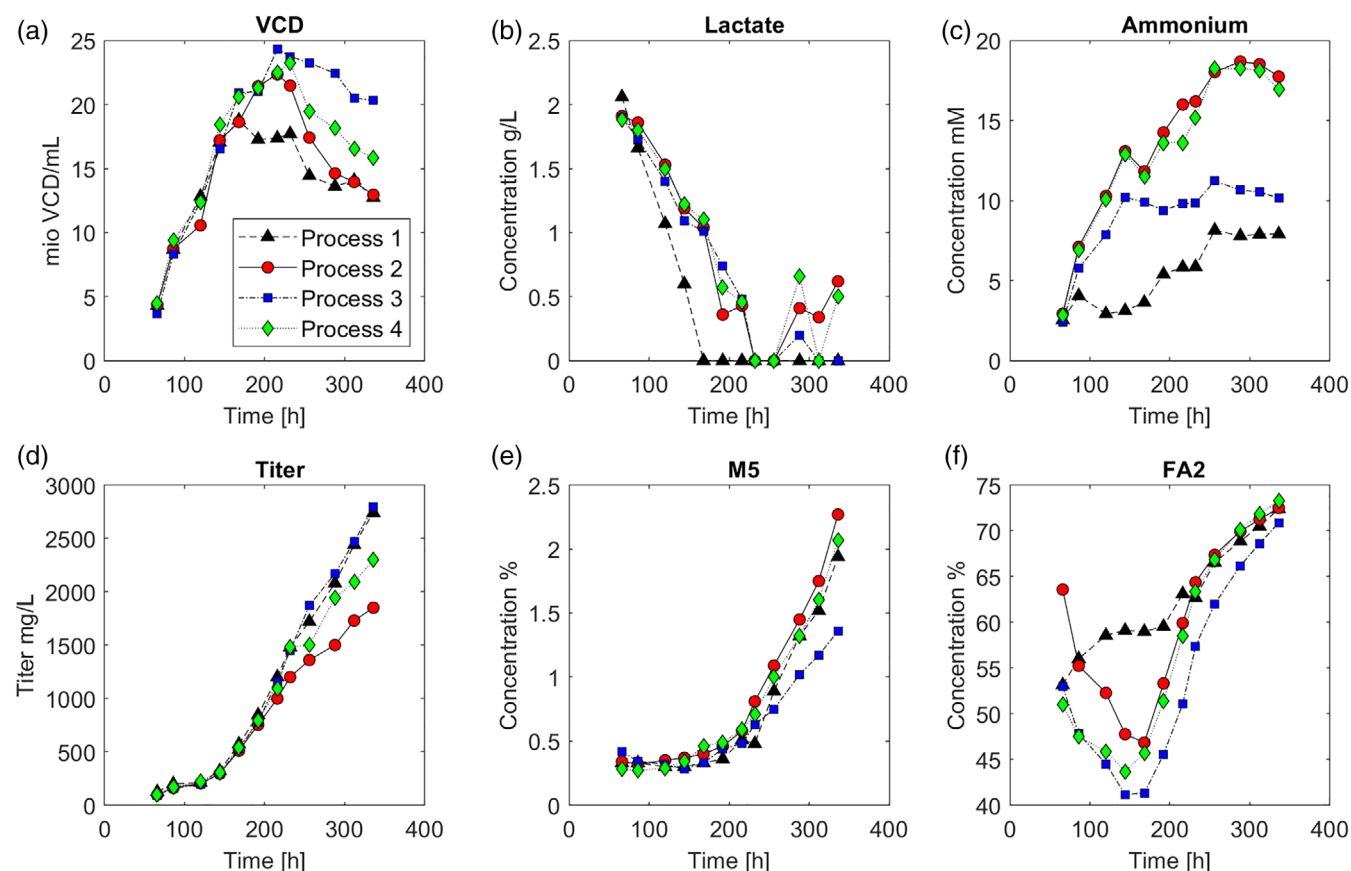


FIGURE 1 Univariate time evolution plots of VCD, lactate, ammonium, titer as well as the glycospecies M5 and fucosylated forms (FA2). Different runs are presented by different markers and colors

cells switched over to a second lactate generation phase at 256 hr. The lowest amount of ammonium was observed in Process 1, reaching 7.9 mM at harvest. Process 3 displayed sustained ammonium production until the 144 hr time point. Subsequently, it reached a plateau in the range of 9.4 to 11.2 mM. Both, similar and high ammonium concentrations, were measured in Processes 2 and 4, peaking between 18 and 19 mM at 288 hr.

The mannose 5 glycan abundance (Man 5) was comparable in all conditions until 225 hr after inoculation (Figure 1). Subsequently, Processes 1–3 were characterized by a substantial increase of the Man5 level, reaching about 2%, while batch 4 exhibited significantly lower levels of Man5. Two patterns of fucosylation trends resulted. The level of fucosylated glycans (here represented as FA2) decreased in Processes 2–4 until 144 or 168 hr after inoculation due to an increase of galactosylation. In the second part of the cultures, fucosylation became more abundant again, reaching similar levels, while galactosylation decreased (not shown). On the other hand, in Process 1, the level of fucosylation steadily increased. It is very tedious and impractical to withdraw information on characteristic dynamic behavior of the four Processes across all considered variables, which suggests a multivariate investigation offering a joint interpretation in a significantly reduced variable space.

3.2 | Glycan correlation analysis with PCA

The main objective of performing a PCA of the 16 Y-variables (Y-PCA) prior to multivariate regression was to identify the degree of correlation among the glycan variables as well as to visualize their multivariate dynamic behavior. Figure 2a presents the obtained loading plot and Figure 2b shows the score plot, where the observations corresponding to one process are connected and labeled with the

process days. In general, the PCA reveals a high degree of correlation with the first two PCs capturing almost 80% of the total glycan variance, whereby PC1 dominates with 60% of variance explained. From the loading plot, the correlation pattern can be interpreted. Lower processed (e.g., M5 and M6) and intermediate glycoforms (e.g., A1 and FA1) in the glycosylation network³⁴ are located at negative values for PC1, while galactosylated forms (e.g., FA2G2-1 and FA2G2) feature positive values of PC1. Sialylated forms (e.g., FA2G2S1) additionally have large values of PC2. In the loading pattern, one can even identify different reactions across species the glycosylation network,³⁵ moving from high mannose forms such as M5 (negative PC1, positive PC2) through the intermediate FA2 (negative PC1 and PC2) toward more mature glycan forms including FA2G1 (positive PC1) and FA2G2 and eventually to FA2G2S1 (positive PC1 and PC2). The fact that several species such as M4 and M7 as well as Hybrid-F do not follow this pattern can be explained by their small abundance close to LOQ. Therefore, one can also identify potentially noisy glycan variables from this pre-analysis.

The score plot in Figure 2b visualizes the multivariate evolution of the 16 glycovariables in two reduced dimensions. Similarly to the univariate time evolution of the fucosylated species (FA2) shown in Figure 1f, one can observe differences of the four processes at culture day 1, which become most pronounced at day 7. Eventually, they begin to converge, while still showing considerable differences at the process end, which can be also observed for the glycoform M5 in Figure 1e. It is important to highlight that this dynamic pattern unifies the evolutions of all glycoforms. Therefore, one can conclude that Processes 2 and 4 are likely to have a similar CQA behavior while Processes 1 and 3 have a different pattern. In fact, Process 1 has, in line with the observations from the loading plot, the smallest abundance in sialylation (low along PC2) while Process 3 has the largest abundance of sialylated forms (located high values of PC2). The

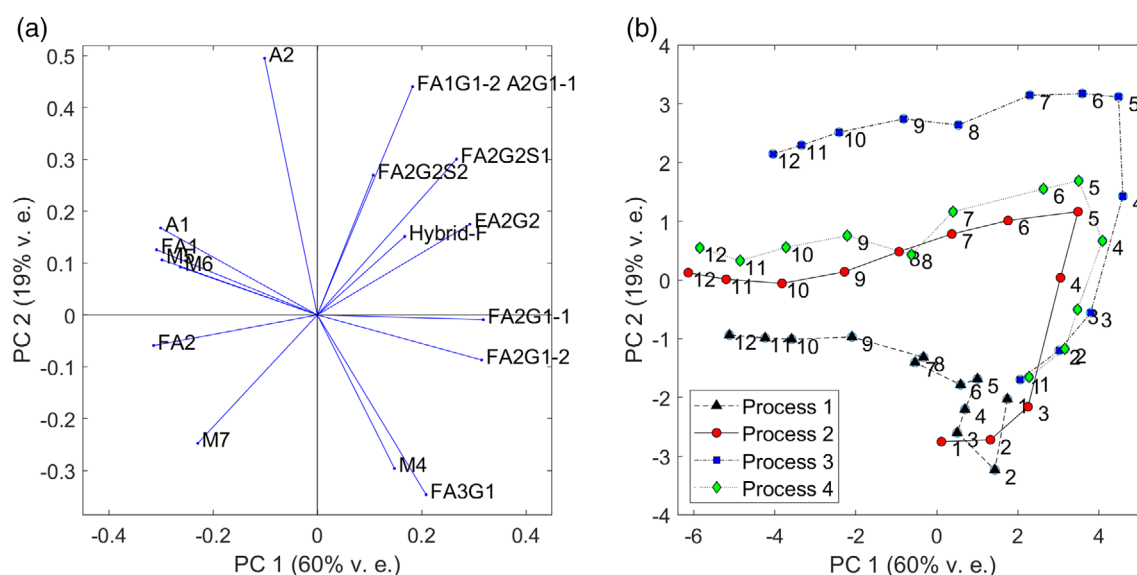


FIGURE 2 PCA of glycosylation variables. (a) Loading plot of the 16 glycoforms; (b) score plot showing multivariate time evolution of the four runs by different colors and markers. The numeric values indicate the process day of the runs

characteristic trajectory of the four processes suggests, in accordance to the loading plot, that over the first days the enzymes active on galactosylation and sialylation are dominant (evolution toward positive PC1 and PC2), while thereafter the processes increasingly evolve toward less galactosylated glycoforms. These effects are also particularly evident in Figure 1f, as the increase of FA2 with ongoing process duration is correlated with increased ammonium level, which inhibit galactosyltransferase resulting in less galactosylated glycoforms being generated. In conclusion, one can highlight that PCA offers practical evaluation of the dynamic evolution of the 16 glycoforms across the four processes in a single plot and reveals their characteristic correlation structure, as seen in Figure 2b.

3.3 | Regression analysis with PLSR

The objective of this analysis is to evaluate the predictability of the glycovariables based on the designed and measured process information, namely the IPC, IPC & Extra, and IPC & Intra variables. Moreover, the importance of specific variables as well as the three aforementioned variable blocks shall be analyzed and compared.

The strong correlation structure observed in the PCA of Y variables (glycovariables) calls for the use of PLS2 models, which regress all of the highly correlated glycoforms based on the process information simultaneously with one single model. This procedure bears the

pronounced advantage of a single model to be interpreted with regards to process variable importance and predictive power. In a first step, PLS2 models are built using the *IPC & Extra* input variables. Figure 3 presents the model quality (Q^2) in calibration (a) and in prediction (b) as boxplots showing the distributions obtained through rotating the process run to be predicted. Thereby, three different modeling procedures for input variable selection are compared, namely using all variables (marked red in Figure 3), those selected by the VIP criterion (marked black) or the GA (marked blue). Comparing the results obtained in calibration (shown in Figure 3a), one can observe a clear trend toward improved values for Q^2 starting from all variables (lowest values), through VIP-based selected variables to GA-based selected variables (highest values). For almost all CQAs, the latter procedure yields very good calibration models with Q^2 above 0.8 as well as narrow distributions indicating model robustness. Three glycoforms, particularly Hybrid-F and M4 but also M7, show rather poor calibration results and little prediction robustness regardless of the chosen selection criteria. This observation is in line with those deduced from the previous PCA. Hence, we decided to exclude these variables from subsequent interpretations. The benefit of variable selection suggests the presence of noise and redundancy among the input variables. The GA is a harsh selection technique eliminating considerably more variables compared to the VIP criterion (70 vs. 50% for VIP). Thereby, it featured a very inconsistent selection patterns across the four predictions (not shown). Moreover, this technique is prone to

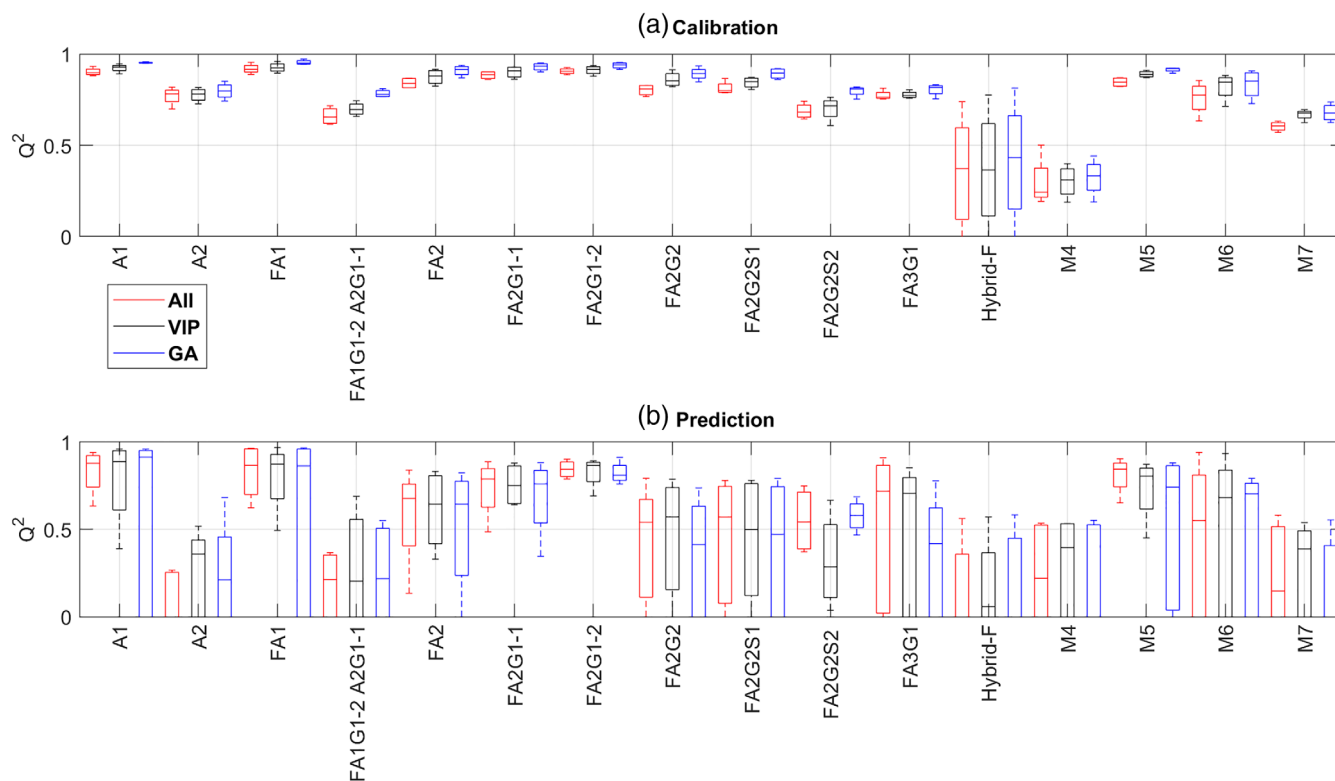


FIGURE 3 PLS2 model results represented as variance explained (Q^2) for 16 CQAs in calibration (a) and prediction (b). The boxplots are based on the four prediction rotations and compare the corresponding results obtained by three variable selection criteria: utilization of all variables (red), VIP-based (black) and GA-based selection (blue)

over-fitting when the selections are made based on a relatively small and therefore limited calibration set. This trend is evident when considering the results for prediction presented in Figure 3b. The GA no longer shows the highest accuracy and, for most of the CQAs features, the largest distributions of predictive accuracy, that is, the lowest model robustness. In other words, given the differences of the process and product quality behavior (cf., Figures 1 and 2b) as well as the limitation of four runs, the GA is biased by the peculiar trends of the three processes used for calibration resulting in poor prediction or extrapolation capability of the forth run. Predictions with all and VIP-selected variables show similar results, but given the complexity reduction offered by the latter, it is considered a better choice, which offers a good compromise between retaining sufficient information for extrapolation and eliminating noise. Therefore, this technique shall be used in all subsequent regression analyses. It is important to point out that the substantial differences between the Q^2 values obtained in calibration and in prediction is likely related to the limited amount of runs available for model calibration. Moreover, the variable-wise unfolding of the data, which was required for such a limited number of runs, bears the assumption that the correlation structure of all variables is constant over time, which introduces further limitations, keeping in mind different metabolic states, such as lactate production and consumption.

In a next step, the benefit of building a model for each CQA individually (PLS1) compared to simultaneous regression (PLS2) is evaluated. Figure 4 presents the Q^2 values obtained in calibration (a) and in prediction (b) when incorporating the *IPC* & *Extra* variable blocks as

input information and using the VIP selection criteria. In calibration, PLS1 models (shown in red) lead to a minor but consistent improvement in the modeling accuracy for most of the CQAs. However, Figure 4b reveals that PLS1 models generally do not outperform PLS2 models and, in some cases, result in less accurate (e.g., for M5) or less robust (e.g., for FA1) predictions (cf., Figure 3b). This is likely to be due to over-fitting based on reducing variables with the VIP criterion. For PLS2 models, such variable elimination is stabilized by the requirement of simultaneously predicting several glycoforms, which imposes the biologically driven CQA correlations. Given this comparison, PLS2 models shall be utilized in the subsequent analysis with the tremendous advantage of having a single process-to-product-model to be calibrated and interpreted.

The added value of the explanatory power of the extracellular variables is now compared to utilizing only the IPC information or utilizing the intracellular information instead. In order to simplify interpretation, this comparison was performed on the five grouped glycoforms, namely the high mannose, afucosylated, sialylated, fucosylated, and galactosylated forms. Figure 5a shows that the first three groups show relatively low abundance, while Figure 5b shows that fucosylated and galactosylated species are present at high levels across the four fed-batch experiments. When considering the accuracy during model calibration in Figure 5c,d, it is evident that models with only the IPC variables as predictors (marked in red) show significantly higher RMSEC for all glycans and considerably worse model robustness, as indicated by the error bars. The calibration results obtained using the intracellular information (presented in blue) are

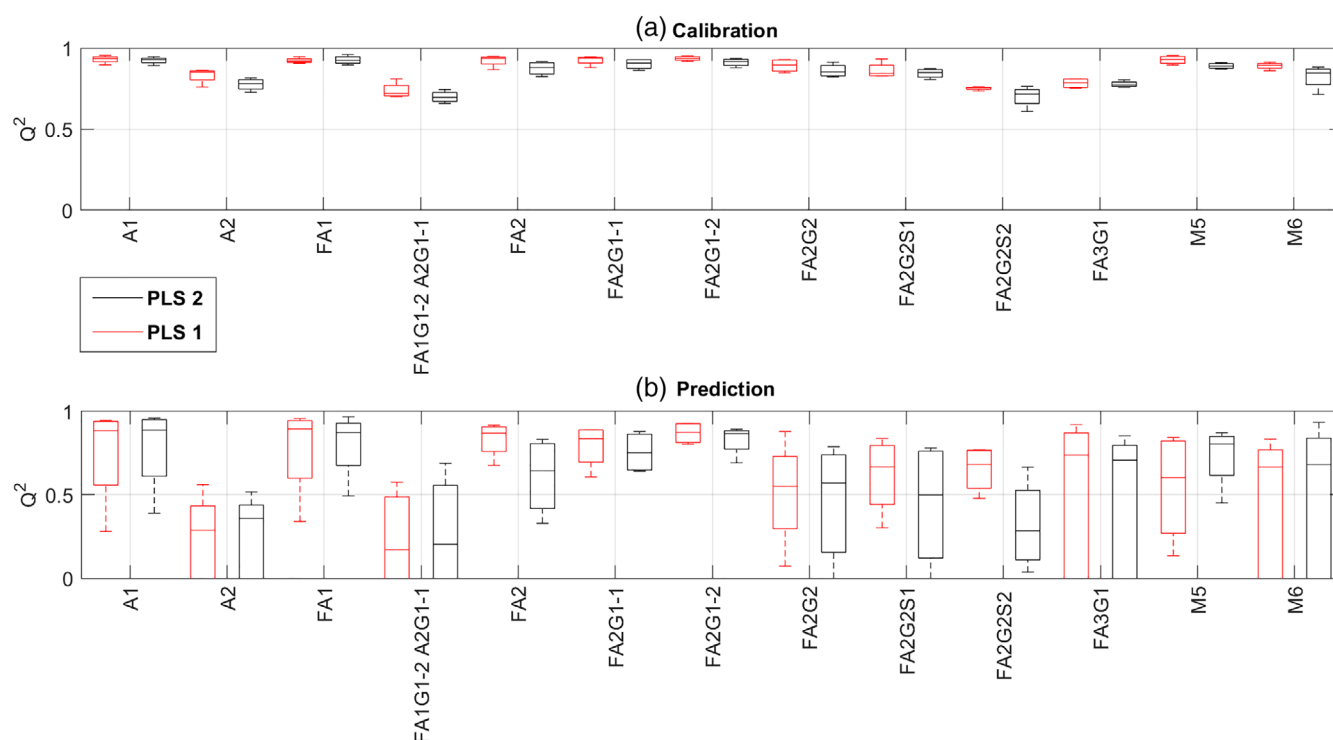


FIGURE 4 Comparison of model performance represented by Q^2 of PLS1 (red) and PLS2 models (black) for 16 CQAs in calibration (a) and prediction (b). The boxplots result from the four prediction rotations. All models were subject to VIP-based variable selection

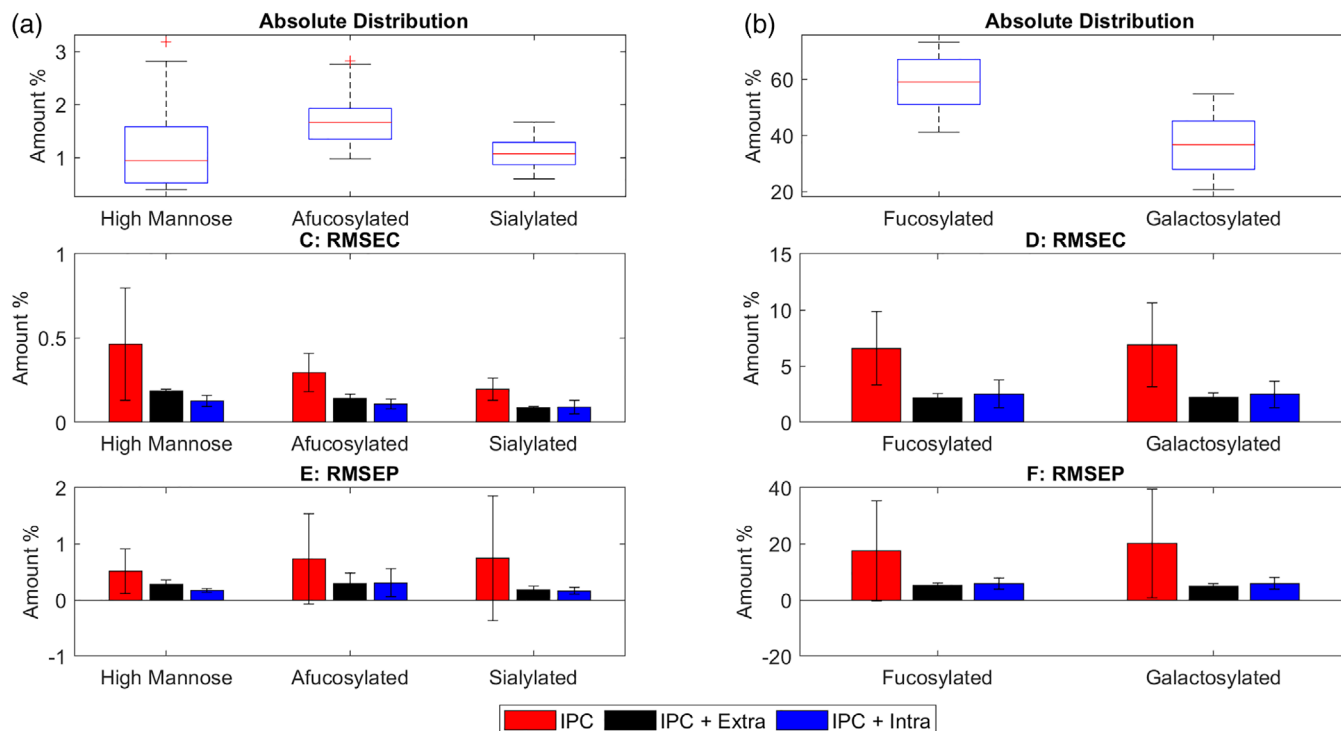


FIGURE 5 Absolute distribution of the five grouped glycoforms (a,b) as well as the corresponding distributions of the RMSEC (c,d) and RMSEP (e,f) in the predictive models. Three cases of different predictor variables are distinguished (*IPC*, *IPC & Extra*, *IPC & Intra*)

very close to the ones based on extracellular metabolomics (in black), with a slight trend to being less robust (particularly for fucosylated and galactosylated glycoforms). This is likely to be a result of the significantly larger number of intracellular variables used in the corresponding linear models, which, given the small number of observations, impose limitations to consistently estimate the PLSR loading structure. On the other hand, one might conclude that the profile of the extracellular metabolome is more homogeneous and sufficiently representative to predict the glycoprofile. It shall be highlighted that for the five glycan groups visualized in Figure 5a,b, the corresponding distributions of the RMSEPs shown in Figure 5e,f are relatively small and indicate that a considerable amount of the process variation can be predicted. The RMSEP distributions show that the combination *IPC&Extra* tends to produce the most accurate and robust prediction results, which are clearly better than using only *IPC*. On the one hand, this demonstrates in more general terms (compared to the calibration results which can be biased by over-fitting) the remarkable additional explanatory power of the extracellular variables. In this case, two- to three-fold accuracy improvement and even more potent robustness improvement can be observed. The corresponding improvement in model robustness is even more significant. The combination *IPC&Extra* with more robust results during calibration also yields much better external predictions than *IPC* only. Keeping in mind the limitation due to the small number of runs, the fact that extracellular, simpler-to-obtain subsets of the generally available metabolomic information can be sufficient is an important observation for future analysis. As already noted for Figure 4, the scarcity of the runs does not allow to

reach further conclusions on the role of extracellular components, although this result remains very encouraging.

The predictions of the different glycoforms with PLS2 models showed considerable variability (i.e., distribution ranges) in Figure 4b and Figure 5 across the four process runs. Moreover, both the univariate evolutions visualized in Figure 1 and the multivariate trajectories from Figure 2b demonstrated considerable differences among the four processes. These observations drove the detailed investigation of the prediction quality obtained for each CQA in the four processes using *IPC & Extra* as input information. Figure 6 shows the corresponding values of Q^2 in prediction, which are marked red for values below 0, that is, denoting lack of statistical significance, and marked with a grey scale for positive values of Q^2 , whereby black elements (with $Q^2 = 0.9-1$) represent almost perfect predictability. When observing the Q^2 values for the different CQAs across the four processes, it can be noted that the predictions of results for Process 1 show significantly smaller values for many variables including A2, FA1G1-2, FA2G2, and FA3G1, that is, almost the complete sequence of the glycosylation network. In fact, the multivariate trajectory of the glycan evolution shown in Figure 2b reveals that this process behaved most differently from the others. One major difference was the absence of manganese in the medium of Process 1, which, particularly in the beginning, promotes the formation of the FA2 species³⁶ as shown in Figure 1f due to reduced galactosylation. Therefore, a model calibrated on the other processes is likely not to be capable of extrapolating toward this behavior. As for the other runs, Process 3 shows weaker prediction of glycoforms including FA1G1-2 and the high

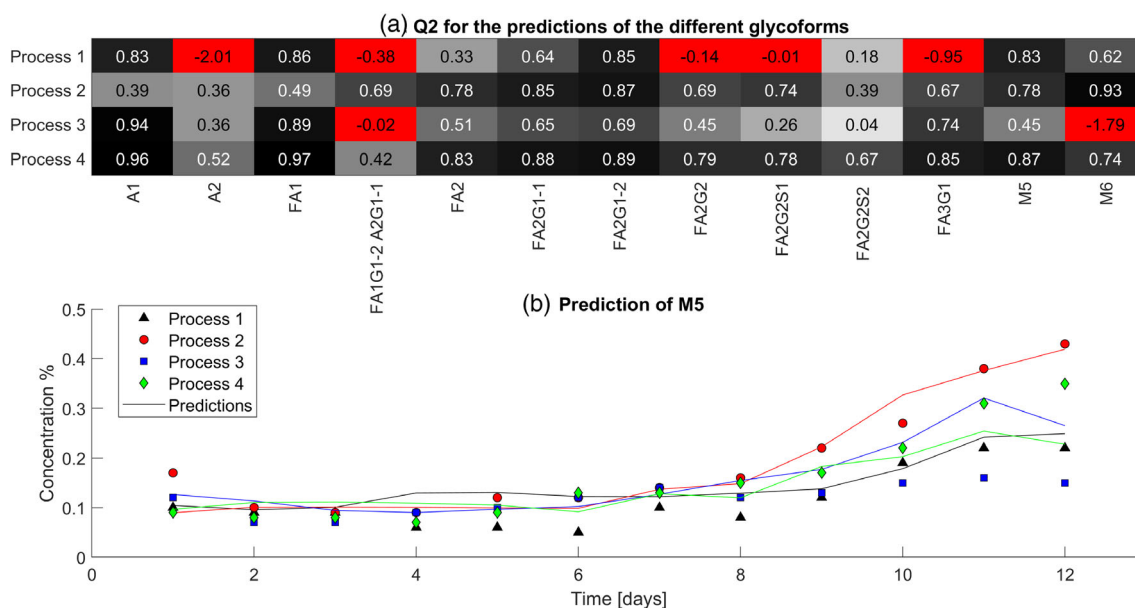


FIGURE 6 (a) Q^2 in prediction (of the corresponding process run) for the 16 glycoforms. Dark fields represent good model performance; red fields indicate unacceptable model performance; (b) Prediction of M5 for four runs (lines) compared to experimental data (symbols)

mannose forms. Again, these observations are in line with the PCA in Figure 2, where the time evolution of this process considerably differs across PC2 (high values in score plot in Figure 2b), where FA1G1-2 and high mannose species show high loadings in Figure 2a. The prediction results are additionally compared to the experimental data in Figure 6b for the glycoform M5. Here one can observe that, along most of the days, the modeling procedure can predict the dynamic behavior of M5 well for all runs with the major exception of Process 3. In that case, it overpredicts the final values showing that M5 extrapolation toward the special behavior of Process 3 cannot be achieved based on training on the more similar evolutions of the other runs. To conclude, one can state that considerable variability has been introduced into the four processes for predictive analysis enabling to build predictive models with an acceptable to good external prediction accuracy for many glycans despite the very small number of observations and absence of replicate experiments.

Heat maps were built based on the VIP selection scheme of the PLS2 models of Figure 6 to support the incorporation of extracellular information in future work. In Figure 7, deep red colors correspond to low values of the VIP (i.e., low importance). Importance increases through black-colored elements (denoting average importance) to deep green-colored elements (for very important metabolites). The four columns represent the VIP pattern obtained when the respective process run (column number) was predicted. One can observe that the general selection pattern (coloring) is similar across all four predictions suggesting that the VIP selections are consistent as described earlier with regards to Figure 3. Moreover, one can highlight that the overall number of considered species could be reduced at least by 50% (VIP ≥ 1 corresponding to all nonred elements), while about 25% of the measured compounds showed consistently very high VIP values (colored in green) in all four predictions.

It is worth commenting about the selection of some of the variables in Figure 7, so to link such statistical analysis to biological considerations. Except for glucose (both as at-line but also as extracellular measurement, showing analytical consistency), all the other five IPC variables appear to be always important. The interrelationship of culture productivity in terms of titer and VCD to the product quality can be also observed in Figure 1, which shows that for high titers the abundance of the intermediate forms is significantly higher,³⁷ while that of high mannose forms is significantly lower. The low VIP of glucose may result from sufficiently high glucose levels throughout the culture. It has been reported that glucose starvation lowered intracellular concentrations of UDP-GlcNAc and UDP-GalNAc,³⁸ which seems not be the case in the four processes. On the other hand, UDP-GlcNAc, the substrate of both GlcNAc transferases (GnT I and GnT II) is an important variable. It plays, in combination with the two enzymes, an important role in the glycan maturation process from Man5 (mannose 5) to complex glycans in the Golgi apparatus in two separate reaction steps; first, in cis-Golgi and second, in medial-Golgi. Interestingly uridine, which is a precursor of UDP and thus UDP-GlcNAc, was not an important variable. This does not necessarily mean that this variable is generally not important, but it is likely that the information provided by it is better or similarly well expressed by other variables (due to detection accuracy, dynamic consistency, etc.), so that such redundancy is eliminated by the VIP selection focusing on unique interdependence patterns. Lactate is known as one of the key metabolites of mammalian cell cultures, whereby particularly high levels of it can have a strong impact on the cell culture process and, thus, also affect glycosylation.³⁹ Ammonium affects the intracellular pH and correspondingly the activity of galactosyltransferase (GalT). High levels of ammonium have been reported to inhibit GalT,⁴⁰ and also to elevate Golgi pH causing GalT to mislocalize.⁴¹ The fact that

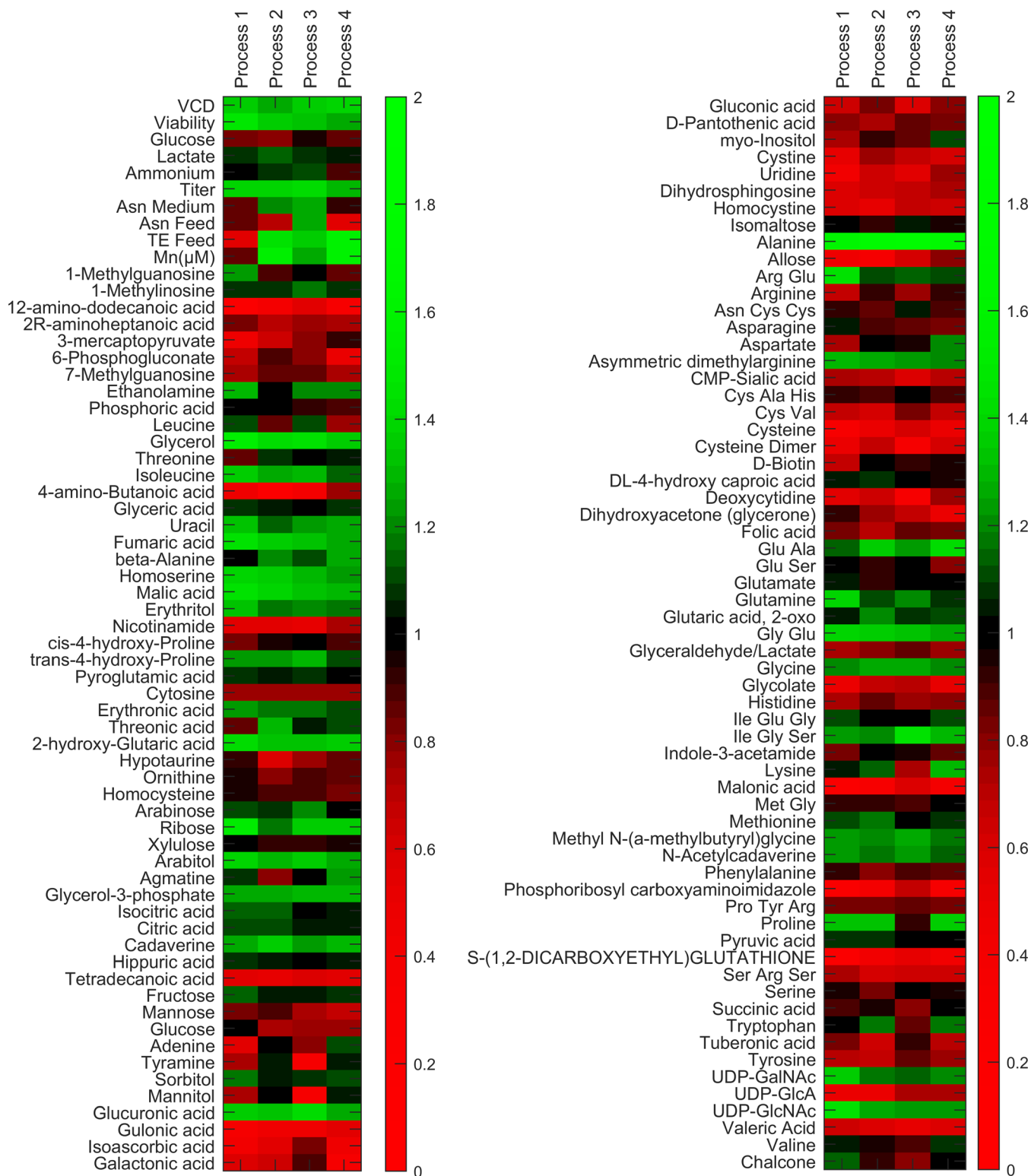


FIGURE 7 Heat Map of the VIP values for four predicted runs (shown in columns) determined from PLS2 models using IPC and extracellular information. Low VIP values are indicated by red fields, neutral values by black and high values by green fields

the highest importance is observed for Processes 2 and 3 can be explained by a substantial decrease of galactosylation in the second half of the culture. The fact that the designed experimental conditions are generally selected except for the prediction of Process 1 confirms that

such induced culture condition changes had an effect on the product quality. Therefore, the conditions for runs 2–4 likely removed significant information from the experimental design for the prediction of process run 1, which was performed under essentially different conditions.

4 | SUMMARY AND CONCLUSION

This work presents the use of a broadly applicable multivariate data analysis methodology tailored to a relatively small cell culture process dataset. The dataset is particularly interesting due to the quantification of hundreds of intracellular and extracellular variables in addition to classical metabolites, such as glucose and lactate. The measured process information was used to predict the dynamic glycosylation behavior and to evaluate the key variables capable of consistently connecting process behavior to the glycopattern of the expressed mAb. The largest challenge was the availability of only four cell culture runs. In order to establish robust predictive models, an approach based on the variable-wise unfolding of the dynamic data (to out-balance the large number of variables with the maximal number of observations, namely 12 per run) was combined with PLS2 models (connecting the process to the product attributes in a single model) and a VIP-based variable selection pattern (to reduce noisy and redundancy among the variables).

The obtained results show that most of the 16 considered glycoforms or five characteristic groups thereof can be accurately predicted for an external run, while calibrating a model with the suggested procedure based on only three runs (respectively, 36 observations). Due to the strong internal correlation of the CQAs, the dynamic process information was regressed to the dynamic glycan pattern with a single model, which significantly simplified the evaluation of the variable importance.

An important result of this work is the fact that the additional metabolomics information remarkably improves the prediction of the glycosylation. It is important to highlight that accurate predictions can be obtained when using standard process and extracellular measurements only, that is, without utilizing the intracellular data. The detailed evaluation of the selected extracellular variables highlighted the central explanatory role of about 25% thereof, whereby many compounds are not known to directly affect glycosylation. Therefore, such model-guided variable reduction could be used to also selectively guide and reduce future analytical effort. In addition, this analysis might direct glycosylation research toward possible additional interrelationships and mechanisms, and most importantly, it can guide cell culture processing toward new monitoring procedures centered on new key compounds to characterize the dynamic process and product quality behavior. All conclusions are certainly limited by the small size of the dataset, but the potential revealed through the suggested simple and widely employable multivariate modeling approach shall drive the integration of an increasing amount of metabolomics data in future bioprocess analysis and development. To overcome the limitations on the number of observations and nonlinearities as well as further guide interpretation, we believe that the exploration of efficient hybrid modeling approaches,⁴² combining tailored statistical tools as presented here with key structures from usually very demanding but effective pathway analysis approaches,⁴³ is likely to be worth exploring for broad application of such quantitative tools.

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